# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — SCIENCE UNIT TEST SET A

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 12 × 1 = 12

*Choose the correct option and write it in your answer book.*

**1.** The Earth spinning on its axis is an example of: 1

**(a)** linear motion **(b)** rotation **(c)** random motion **(d)** oscillation

**2.** The motion of the Earth around the Sun is: 1

**(a)** linear **(b)** periodic (circular/orbital)
**(c)** random **(d)** none

**3.** A car moving along a straight road shows: 1

**(a)** rotation **(b)** random motion **(c)** circular motion **(d)** linear motion

**4.** The property of a material to resist being scratched or dented is: 1

**(a)** transparency **(b)** hardness **(c)** lustre **(d)** solubility

**5.** The ability of a substance to dissolve in water is its: 1

**(a)** solubility **(b)** lustre **(c)** texture **(d)** hardness

**6.** Which is made of a natural material? 1

**(a)** Cotton shirt **(b)** Nylon rope **(c)** Synthetic cloth **(d)** Plastic bottle

**7.** Heat is a form of: 1

**(a)** mass **(b)** force **(c)** energy **(d)** matter

**8.** A thermometer works because the liquid in it: 1

**(a)** shrinks when heated **(b)** expands when heated
**(c)** freezes **(d)** evaporates

**9.** The liquid commonly used in a thermometer is: 1

**(a)** mercury or coloured alcohol **(b)** milk
**(c)** water **(d)** oil only

**10.** Evaporation is faster when the surface area is: 1

**(a)** larger **(b)** zero **(c)** smaller **(d)** constant only

**11.** The process by which water changes to vapour at any temperature is: 1

**(a)** melting **(b)** freezing **(c)** evaporation **(d)** condensation

**12.** The temperature at which ice melts is: 1

**(a)** 100 °C **(b)** −10 °C **(c)** 0 °C **(d)** 10 °C

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 10–15 words each.*

**13.** Write the SI unit of length and one multiple. 2

**14.** What is solubility? Give one soluble and one insoluble substance. 2

**15.** Why do some objects float on water? 2

**16.** Why should the bulb of a laboratory thermometer not touch the bottom of the beaker? 2

**17.** What is freezing? Give one example. 2

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 30–40 words each.*

**18.** Is a passenger sitting in a moving bus at rest or in motion? Explain with references. 3

**19.** What is hardness? How can you compare hardness of two materials? 3

**20.** Write the steps to measure the body temperature with a clinical thermometer. 3

**21.** Explain the factors affecting evaporation. 3

**22.** Explain with a diagram the water cycle. 3

OR

State what happens when water is cooled below 0 °C and when ice is heated above 0 °C. 3

**23.** Distinguish between rotational and oscillatory motion with examples. 3

OR

Give three examples of each: periodic motion and random motion. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 80–100 words each.*

**24.** (i) Give the uses of metals. 5
(ii) Explain malleability and ductility.
(iii) Give examples.

OR

(i) How do we group materials? 5
(ii) Group: glass, brick, cloth, rubber, steel, leather.
(iii) State the basis of grouping.

**25.** (i) Differentiate between evaporation and boiling. 5
(ii) Between melting and freezing.
(iii) Give examples.

OR

(i) Explain the three states of water. 5
(ii) Name the processes that change one to another.
(iii) Draw a flow diagram.

## END OF QUESTION PAPER — Total: 50 Marks